```
xphd
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 \ V /| |__) | |__ | |  | |
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Version 1.1.0  --  Exciton-Phonon Dynamics
Rishabh Saraswat, Mantasha Ansari, Rekha Verma and Sitangshu Bhattacharya
Indian Institute of Information Technology, Allahabad

If you use xphd in published work, please cite:
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  - xphd, Zenodo, https://zenodo.org/records/23232733

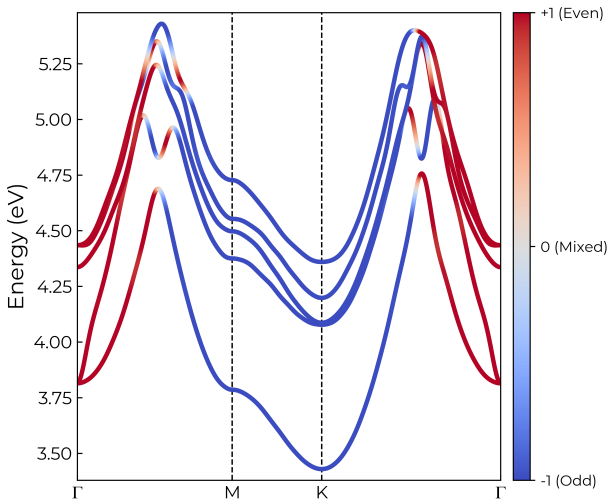
Quantum ESPRESSO + LetzElPhC + yambo  ->  linewidths, symmetry, optics, transport
```

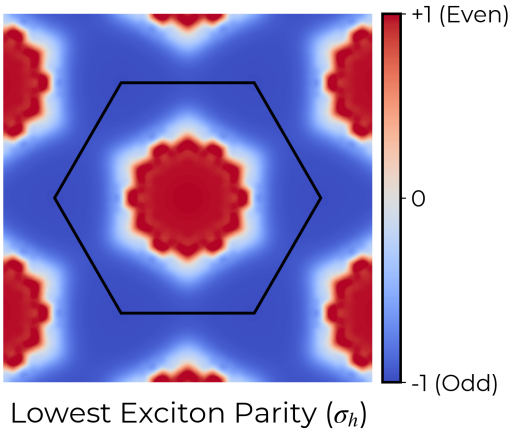
Tutorial: monolayer GaN

from Quantum ESPRESSO to real-time exciton dynamics

This tutorial takes one material through the whole workflow: planar monolayer GaN, whose lowest exciton sits at the corner $\mathbf{K}$ of the Brillouin zone and is therefore dark. Every step gives the command, the files it reads and writes, and *what you should see*. The printed output blocks are copied from our own runs, and the numbers in the green boxes are the ones we obtained, so you can tell at each step whether your calculation is on track before you spend time on the next one.

The methods behind each step (the matrix elements, mirror parity, the triangle integration and the transport equation) are derived in the companion document, `docs/xphd_guide.pdf`. Running on a cluster is covered in `docs/HPC.md`.





What we will find. Left: the GaN excitons on the 24×24 mesh, each point coloured by its mirror parity $\chi(\sigma_h)$. The lowest exciton is at $\mathbf{K}$ (3.429 eV, odd); the bright doublet is at Γ (3.814 eV, even). Right: the parity of the lowest exciton band over the zone: even (red) inside a closed contour around Γ, odd (blue) everywhere else. Most of the tutorial follows from this one picture.

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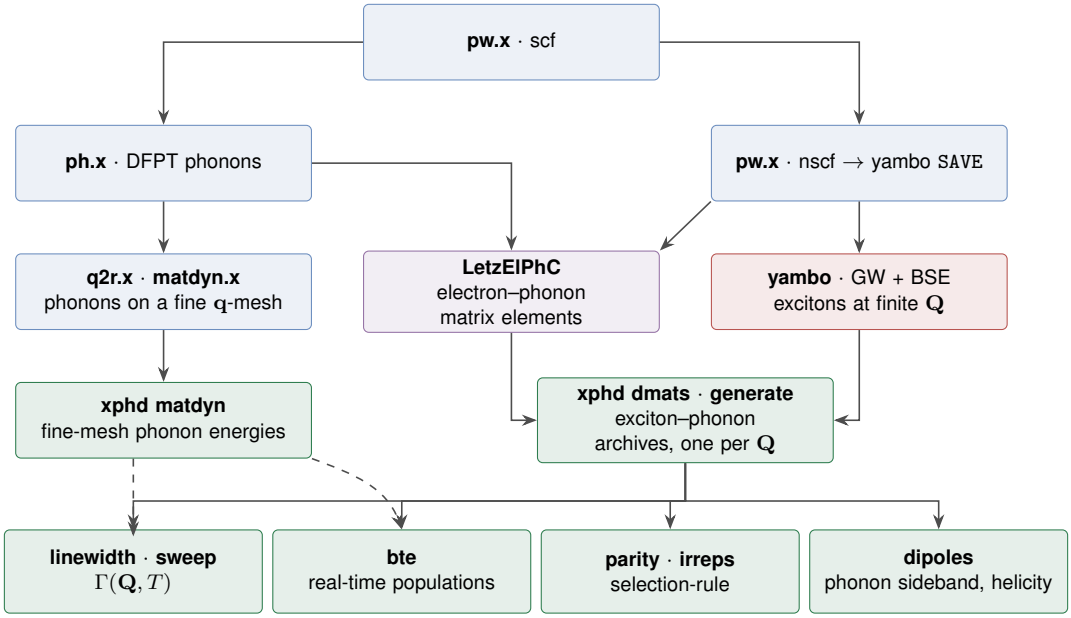
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1. The workflow at a glance



Blue: Quantum ESPRESSO. Red: yambo. Purple: LetzElPhC. Green: `xphd`. Dashed: the fine-mesh phonons, used by the linewidths and the transport. The four analyses in the bottom row are independent of each other once the archives exist. The same chart is in the README.

The system and the parameters we used

Item	Value
Structure	planar honeycomb, Ga and N in one plane; point group D_{3h} (space group $P\bar{6}m2$, No. 187)
Lattice	$a = 3.21$ Å (`celldm(1)` = 6.15 Bohr), 15 Å of vacuum
Ground state	PBE; fully relativistic pseudopotentials with `lspinorb=.false.` (spinless); `ecutwfc` = 80 Ry; $24 \times 24 \times 1$ **k**-points
Phonons	DFPT on $24 \times 24 \times 1$, `tr2_ph` = 10^{-18}; 6 branches; `loto_2d`, crystal acoustic sum rule
Fine mesh	360×360 (refinement factor 15), from `matdyn.x`
GW	G_0W_0 with 100 bands (9 occupied + 91 empty)
BSE	bands 7–12 (3 valence + 3 conduction, `-nv 3`), 15 states at each of the 61 irreducible **Q**
Archives	576, one per **Q** of the full zone; `GI_ExcPh_Q0001.npz` is Γ, `GI_ExcPh_Q0575.npz` is **K**

Directory layout

Keep one directory per stage. The nscf overwrites `<prefix>.save`, and `ph.x` must read the scf ground state, so each stage gets its own copy of the scf save.

```
GaN/
  01_scf/          pw.x scf (and the vc-relax before it)
  02_phonons/      ph.x, q2r.x, matdyn.x  (360x360 chunks)
  03_nscf/         pw.x nscf  ->  p2y  ->  SAVE/
  04_gw/           yambo G0W0
  05_bse/          yambo BSE at every irreducible Q  ->  ndb.BS_diago_Q1 ... Q61
  06_lelph/        LetzElPhC  ->  ndb.elph
  07_xphd/         Dmats.npy, archives/, hw_fine.npy, everything that follows
```

Install

```
git clone <repository> xphd && cd xphd
pip install -e ".[generate]"        # numpy, netCDF4, tqdm; install yambopy from its source
pip install scipy                   # the transport solver
xphd --version                      # xphd 1.1.0
pytest                              # synthetic data only, no databases needed
```

On a cluster, build `mpi4py` against the cluster's MPI first and add the `hpc` extra: `scripts/hpc/install_cluster.sh` does both, and `docs/HPC.md` explains why.

2. Ground state

2.1 Relax, then check that the layer is planar

We started the variable-cell relaxation (`calculation='vc-relax'`, `cell_dofree='2Dxy'`, `forc_conv_thr` $= 10^{-5}$ Ry/Bohr) from a buckled cell with a 0.869 Å height difference between Ga and N. It converged to a *planar* layer: the residual height difference is 9.3×10^{-6} Å, far below what the force threshold can resolve.

Symmetry tolerance. QE, VESTA and LetzElPhC reported C_{3v} for this cell because their tolerances (10^{-5}–10^{-6}) are tighter than the residue. spglib at `symprec` $= 10^{-4}$ Å correctly reports D_{3h}. Rerun everything on the symmetrised planar cell (`xphd validate-planar` lists the checks), so that nothing in the calculation mixes two geometries.

2.2 scf

01_scf/scf.in

```
&CONTROL
  calculation   = 'scf'
  prefix        = 'gan'
  pseudo_dir    = './'
  verbosity     = 'high'
  wf_collect    = .true.
  etot_conv_thr = 1e-05
  forc_conv_thr = 1e-05
/
&SYSTEM
  ibrav = 0, celldm(1) = 6.15, celldm(3) = 4.61
  nat = 2, ntyp = 2
  ecutwfc = 80
  occupations = 'fixed'
  force_symmorphic = .true.
/
&ELECTRONS
  diagonalization = 'cg', diago_cg_maxiter = 80
  conv_thr = 1e-14, electron_maxstep = 400
/
ATOMIC_SPECIES
  Ga  69.7230  Ga.upf
  N   14.0067  N.upf
CELL_PARAMETERS (alat = 6.15000000)
   0.987484665   0.000000000   0.000000000
  -0.493742332   0.855186805   0.000000000
   0.000000000   0.000000000   4.610000000
ATOMIC_POSITIONS (alat)
N    0.0000000000   0.0000000000   0.0000000000
Ga   0.4937423323   0.2850622742   0.0000000000
K_POINTS {automatic}
  24 24 1 0 0 0
```

```
mpirun -np 48 pw.x -in scf.in > scf.out
```

What you should see

- 12 spatial symmetry operations (D_{3h}; a buckled cell gives 6, C_{3v}). yambo and LetzElPhC later report 24, the 12 times time reversal.
- $a = 6.15 \times 0.987485$ Bohr $= 3.214$ Å; the vacuum is 4.61×6.15 Bohr $= 15.0$ Å. Pass the same lattice constant (`-alat 3.21`, `-hex 3.21`) to every analysis script that converts to Cartesian units.
- The irreducible wedge holds 61 of the 576 points. σ_h acts trivially on in-plane wavevectors, so the count is the same for C_{3v}; what D_{3h} saves is in the number of irreducible representations `ph.x` handles, not in the $\mathbf{q}$-list.

3. Phonons

3.1 DFPT on the 24×24 mesh

Take the $\mathbf{q}$-mesh equal to the $\mathbf{k}$-mesh, so that every $\mathbf{k}+\mathbf{q}$ lies on the mesh. The settings that matter for what follows are the threshold and the potential perturbations, which LetzElPhC reads:

02_phonons/ph.in (the settings that matter here)

```
phonons
&inputph
  prefix       = 'gan'
  fildyn       = 'gan.dyn'
  fildvscf     = 'dvscf'
  tr2_ph       = 1.0d-18
  alpha_mix(1) = 0.7
  ldisp        = .true.
  nq1 = 24, nq2 = 24, nq3 = 1
/
```

`tr2_ph` is a threshold on the *squared* change of the potential, so its unit is Ry^2. With images (`ph.x -ni N`) every image works in its own `_ph{i}` directory, so `dvscf` files and patterns are spread over them; gather them before LetzElPhC (Sec. 5). A restart (`recover=.true.`) must keep the same process and pool layout.

3.2 Force constants and the 360×360 mesh

q2r.in and matdyn.in (namelist only; the q-list is appended per chunk)

```
&input
  fildyn  = 'gan.dyn'
  zasr    = 'crystal'
  flfrc   = 'gan.fc'
  loto_2d = .true.
/
```

```
&input
  asr             = 'crystal'
  flfrc           = 'gan.fc'
  flfrq           = 'gan.freq'
  flvec           = 'matdyn.modes'
  loto_2d         = .true.
  q_in_cryst_coord = .true.
  q_in_band_form  = .false.
  nosym           = .true.
/
```

`matdyn.x` is serial and the fine mesh has 129 600 points, so the list is split into chunks that run side by side, and the outputs are merged. None of `matdyn`'s output files survives a plain `cat`: each chunk carries its own header and its own cumulative path distance.

```
xphd matdyn-split --n 360 --chunks 54                 # qlist_000.txt ... qlist_053.txt
sbatch scripts/run_matdyn.sh                          # runs every chunk, then merges
xphd modes-check  --modes matdyn.modes --freq gan.freq --n 360
xphd matdyn gan.freq --fine 360 --nbnd 6 --npz archives/GI_ExcPh_Q0001.npz -o hw_fine.npy
```

The last command needs the Γ archive (Sec. 6) only to compare the two meshes where they coincide; it can wait until then.

What you should see

- `modes-check` should report 129 600 $\mathbf{q}$-blocks and every mode planar: purely in-plane or purely out-of-plane ($|e_z|^2$ is 1 or below 3×10^{-4}).
- `xphd matdyn`: the fine-mesh frequencies reproduce the archive's `hw_grid` to 1.7 μeV on every branch. The only exception is the acoustic residue at Γ, 0.237 meV in the archive (which carries the DFPT frequencies at $\mathbf{q}=0$) and exactly zero in `hw_fine.npy`, which is what the linewidths and the transport use.
- Phonon energies from 0 to 101.5 meV.

`loto_2d=.false.` does *not* switch the non-analytic term off; it selects the three-dimensional form, which puts a spurious ≈15 meV LO–TO splitting at Γ. Use `loto_2d=.true.` in both `q2r.x` and `matdyn.x`.

4. Quasiparticles and excitons

4.1 nscf and the yambo SAVE

Run a non-self-consistent calculation on the same $24 \times 24 \times 1$ mesh with enough empty bands for the GW (we used 100), then convert it with p2y and initialise with yambo. This SAVE is used by the GW, the BSE *and* LetzElPhC.

One SAVE for everything. The exciton–phonon matrix elements combine the BSE eigenvectors and the electron–phonon elements in one wavefunction gauge. If you redo the nscf, redo both the BSE and LetzElPhC.

4.2 G_0W_0

For a two-dimensional layer use the slab cutoff (`CUTGeo= "slab z"`) with the random-integration method, and the Bruneval–Gonze terminator (`GTermKind="BG"`, `XTermKind="BG"`) to speed up the convergence in bands.

What you should see

G_0W_0 gap 4.78 eV, indirect ($\mathbf{K} \to \Gamma$); direct gap 4.91 eV. The valence maximum at $\mathbf{K}$ is N p_z (odd under σ_h); the conduction minimum at Γ is Ga s (even).

4.3 BSE at every irreducible $\mathbf{Q}$

- `BSEBands` = 7–12: three valence and three conduction bands, 9 transitions per $\mathbf{k}$ and a matrix of $576 \times 9 = 5\,184$.
- `BSEQptR` over all 61 irreducible $\mathbf{Q}$, which writes `ndb.BS_diago_Q1` … `Q61`.
- Keep at least the 15 lowest states at every $\mathbf{Q}$.
- Check that the report's `[K+QP] Direct Gap` equals the `[X]` gap plus the GW correction, so you know the quasiparticle shift was applied.

What you should see

	energy (eV)	$\chi(\sigma_h)$
lowest exciton, $\mathbf{K}$ (and $\mathbf{K}'$)	3.4290	-1 (A″, dark)
bright doublet, Γ	3.8143, 3.8171	$+1$, $+1$ (E′)
lowest exciton, $\mathbf{M}$	3.7856	-1

The bright exciton lies 385 meV above the dark one. The 2.8 meV between the two members of the bright doublet is the long-range exchange at $\mathbf{Q} = 0$, evaluated along one direction; neither member alone is C_3-symmetric, only the pair is.

Planar versus buckled. On the same mesh the buckled cell gives 3.3569 eV at $\mathbf{K}$ and 3.7358 eV at Γ, both lower by 72–79 meV: a rigid shift. The difference $\Gamma - \mathbf{K}$ moves by only 6 meV, so the energy-conserving delta functions, and with them the linewidths and the transport, are unchanged. An exciton band file whose minimum is near 3.355 eV belongs to the buckled cell.

5. Electron–phonon matrix elements: LetzElPhC

- Gather the phonon run into one `ph.save` directory: `dyn0...dynN`, `dvscf1...dvscfN`, `patterns.N.xml`, `tensors.xml`, `data-file-schema.xml` and the pseudopotentials, from every `_ph*` image. A $\mathbf{q}$-point split between two images leaves partial `dvscf` files; check that each $\mathbf{q}$'s file has the same size in every image.
- In `lelphc.in` set `ph_save_dir = ph.save`. The key is case-sensitive and a misspelling is silently ignored; the symptom is `Unable to open the dyn0 file`.
- LetzElPhC removes the arbitrary phases of the single-particle states and writes `ndb.elph`, with the electron–phonon elements on the full 24×24 mesh of $\mathbf{k}$ and $\mathbf{q}$.

6. Exciton–phonon archives

An *archive* holds, for one exciton momentum $\mathbf{Q}$, the matrix elements $\mathcal{G}_{\beta\lambda\nu}(\mathbf{Q}, \mathbf{q})$ to every $\mathbf{q}$ of the mesh, every branch ν and every pair of states, together with the energies it needs. Everything after this section reads archives only.

6.1 Rotation matrices

```
cd 07_xphd
xphd dmats --bse-dir ../05_bse/output --savepath ../03_nscf/SAVE --ndb-elph ../06_lelph/ndb.elph --nexc 15
```

`Dmats.npy` holds the single-particle rotation matrices for every symmetry operation and $\mathbf{k}$, in the BSE band window. They rotate the excitons computed in the irreducible wedge onto the full zone. They depend on the `SAVE` and the band window, not on the BSE solution, so they stay valid if only the BSE is redone.

6.2 The Γ archive first

```
xphd generate 0 archives --nexc 15 --mesh 24 24 --savepath ../03_nscf/SAVE \
      --ndb-elph ../06_lelph/ndb.elph --bse-dir ../05_bse/output --dmats Dmats.npy
xphd check archives/GI_ExcPh_Q0001.npz
xphd check-archive archives/GI_ExcPh_Q0001.npz
```

output (xphd check)

```
integrity checks
  |G|^2 vs g2              0.00e+00   OK
  Ge + Gh vs G             1.24e-15   OK
  E_n vs E_m(q=0)        0.0000 meV   OK
  acoustic |G|^2 at Gamma  0.00e+00   OK
  |G| median 0.001 meV, max 807.9 meV
```

output (xphd check-archive, last two sections)

```
  CONSISTENCY
   mesh 24x24, 6 branches, 15 states, Q = [0. 0. 0.]
   OK  |G|^2 == g2        0.00e+00  (tol 1e-10)
   OK  Ge + Gh == G       1.24e-15  (tol 1e-05)
   OK  E_m(q=0) == E_n    0.00e+00  (tol 1e-06)
   acoustic |G|^2 at Gamma: 0.00e+00 eV^2  (should vanish by the acoustic sum rule)
   phonons 0.00 .. 101.52 meV

  SYMMETRY
   OK   time reversal  q -> -q               L1 1.3e-03  max 2.1e-03   (energies differ by 0e+00 eV)
   OK   C3 rotation    q -> C3 q (60 deg basis) L1 3.2e-03  max 7.8e-03   (energies differ by 0e+00 eV)
   |G|^2 summed over manifolds closer than 15 meV (7 initial groups); OK below 0.01, FAIL above 0.05.

   576 q-points on a 24x24 mesh -> generate iQ = 0 .. 575 for the full set of 576 archives
   READY: launch the rest.
```

What you should see

- The electron and hole parts add up to the total to 10^{-15}, and the acoustic coupling at Γ vanishes *exactly*. That second identity fixes the relative sign of the electron and hole terms: with the wrong sign they add at $\mathbf{q} = 0$ instead of cancelling.
- Time reversal and C_3 both hold at the 10^{-3} level. One operation far above the other points to a rotation fault, unless `-detail` shows a split doublet (then raise `-deg-tol`).
- **READY**. Do not generate the other 575 archives before you see it.

6.3 The rest of the zone

The transport connects states at different $\mathbf{Q}$, so it needs *every* archive; a missing $\mathbf{Q}$ would act as a perfect sink. The linewidths alone need only the 61 irreducible ones (`xphd ibz` lists them), but generating all 576 costs little more and lets every later step use the same set.

```
# one machine: the databases are loaded once, finished archives are skipped
xphd generate 1-575 archives --resume --nexc 15 --mesh 24 24 --savepath ../03_nscf/SAVE \
      --ndb-elph ../06_lelph/ndb.elph --bse-dir ../05_bse/output --dmats Dmats.npy
# a cluster, over MPI ranks (edit the paths at the top of the script)
sbatch scripts/hpc/slurm_generate.sh
xphd verify-archives --dir archives --n 576
```

What you should see

`verify-archives`: 576 present and complete. If a job stops at the time limit, submit it again unchanged; complete archives are skipped.

7. Linewidths

The linewidth is the golden-rule rate of scattering out of a state by emitting or absorbing one phonon. The energy-conserving delta function is integrated exactly over the triangles of the 360×360 mesh, with no broadening parameter: exciton energies are refined by local quadratics around each mesh point, and the phonons come from `hw_fine.npy`.

7.1 The bright exciton at Γ

```
xphd linewidth archives/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 5 77 300 \
      --acoustic-cut 5e-4 --acoustic-model q2 --n-shell 1 --per-mode -o G.txt
```

What you should see

State 1 (the bright exciton): 36.7 meV at 5 K (99.8% emission), 51.9 meV at 77 K (lifetime $\hbar/\Gamma = 12.7$ fs), 150.9 meV at 300 K (4.4 fs; 43% absorption). State 2: 56.8 meV at 77 K.

The acoustic model. By the acoustic sum rule the constant-coupling value in the mesh cell around $\mathbf{q} = 0$ is zero. `-acoustic-model q2` replaces it there by the analytic form $|\mathcal{G}|^2\, 2\hbar\omega = Cq^2$ fitted to the neighbouring shells. Without it (the default, `none`) the same state reads 46.8 meV at 77 K. Use the same model in the single-$\mathbf{Q}$ runs and in the sweep, or a minimum will read differently in the zone maps and in the temperature plots.

7.2 Where a width comes from

```
xphd channels archives/GI_ExcPh_Q0001.npz --state 1 --T 77 --acoustic-cut 5e-4
```

output (on the 24x24 mesh, no refinement)

```
State 1 at 77.0 K: Gamma = 46.812 meV over 13 live channels

      ch  nu   m     <hw>   DOS 1/eV  max|G| meV        meV   % tot
    emis   1   1     8.66      1.281      94.079     33.152   70.8%
    emis   3   1    24.89      1.328      99.549      6.846   14.6%
     abs   1   1     8.66      1.036      94.079      3.585    7.7%
    emis   2   1    17.67      1.313      67.436      2.910    6.2%
     abs   3   1    24.89      1.007      99.549      0.271    0.6%
     abs   2   1    17.67      1.018      67.436      0.048    0.1%
     abs   5   1    91.77      0.953     137.634      0.000    0.0%
     abs   6   1   100.34      0.925     178.344      0.000    0.0%
     ...
```

What you should see

The three acoustic branches carry the whole width. The polar optical branches 5 and 6 have the largest matrix elements of all (138 and 178 meV) and contribute nothing: every final state they could reach in the lowest band is odd, and the bright exciton is even. Section 8 makes this quantitative.

7.3 The dark minimum at K

```
xphd linewidth archives/GI_ExcPh_Q0575.npz --hw-fine hw_fine.npy --T 5 77 300 \
      --acoustic-cut 5e-4 --acoustic-model q2 --n-shell 1 --per-mode -o K.txt
python scripts/check_minimum_linewidth.py        # SETTINGS: ARCHIVE, ALAT, MASSES
```

What you should see

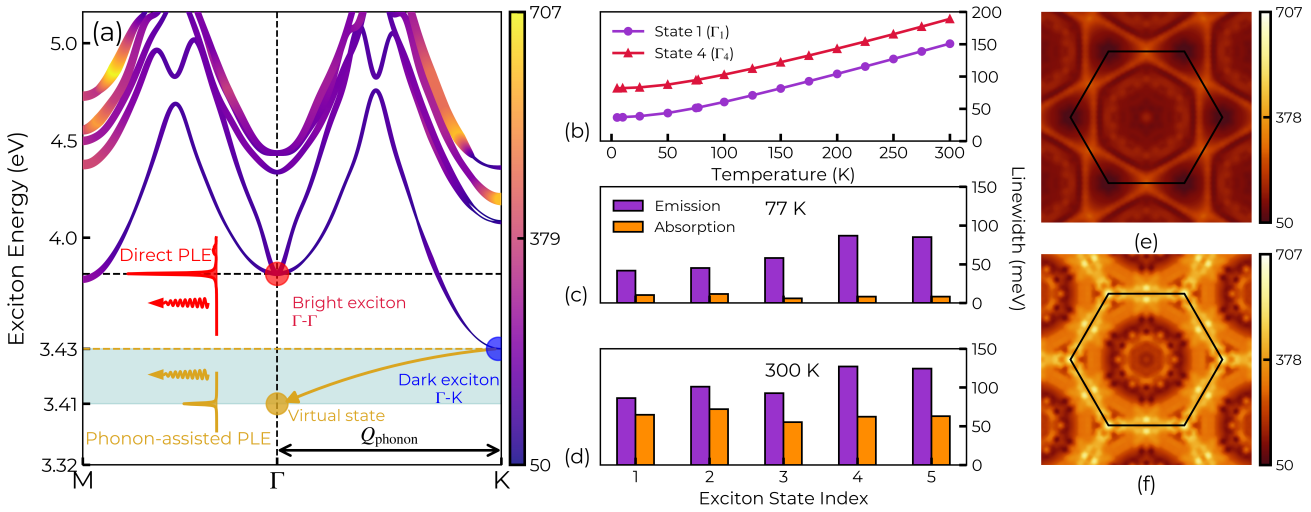
- With q2: 0.087, 10.4 and 50.5 meV at 5, 77 and 300 K. This is quasi-elastic LA absorption inside the valley.
- With the default model the width is about 0.04 meV at 77 K, i.e. ≈ 15 ps: the rate of leaving the valley bottom.

- The closed form $\Gamma = (4\pi^2/\sqrt{3})\, C\, N(\hbar\omega^*)/b^2$, with the curvature $b = 12.44$ eV $|\mathbf{b}|^{-2}$ and $\hbar\omega^* = 1.49$ meV, gives 10.6–11.4 meV at 77 K against the code's 10.4 meV.

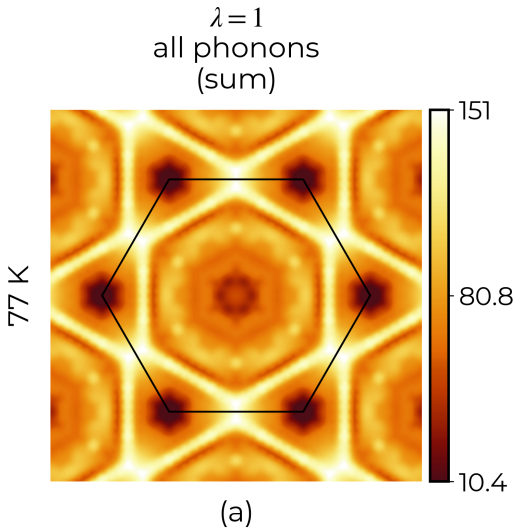
7.4 Over the whole zone, every temperature at once

```
xphd sweep 'archives/GI_ExcPh_Q*.npz' --hw-fine hw_fine.npy --T 5 10 20 40 77 100 150 200 250 300 \
      --acoustic-cut 5e-4 --acoustic-model q2 --n-shell 1 --workers 8 \
      --unfold ../03_nscf/SAVE --checkpoint ck_sweep -o lw_field.npz
```

The delta-function weights do not depend on temperature, so a sweep with ten temperatures costs about as much as one with a single temperature. `lw_field.npz` holds `LW` (temperature, $\mathbf{Q}$, state) in eV, the energies, and the emission and absorption of every branch (`mode_em`, `mode_ab`). On a cluster use `scripts/hpc/slurm_sweep.sh`.



(a) The exciton bands, with the 300 K width as line thickness and colour, and the two luminescence channels: direct from the bright exciton at Γ, and phonon-assisted from the dark exciton at $\mathbf{K}$. (b) Width of states 1 and 4 at Γ from 5 to 300 K. (c,d) Emission and absorption of the first five states at Γ at 77 and 300 K. (e,f) The 300 K width of states 1 and 4 over the zone.



Width of the lowest exciton band over the zone at 77 K, from `lw_field.npz` (default acoustic model). It is smallest at $\mathbf{K}$, where the dark minimum has almost nowhere to go: at 77 K it can only absorb low-energy acoustic phonons.

8. Symmetry: mirror parity and the selection rule

The horizontal mirror σ_h leaves every in-plane wavevector unchanged, so every electron, phonon and exciton state has a parity $p = \pm 1$ *at every point of the zone*, not only at the high-symmetry points. A phonon ν can scatter an exciton λ into β only if

$$p_\beta\, p_\nu\, p_\lambda = +1 :$$

even phonons connect excitons of equal parity, odd phonons excitons of opposite parity.

8.1 Exciton parity across the zone

```
xphd parity --save ../03_nscf/SAVE --bse-dir ../05_bse/output --dmats Dmats.npy \
      --nv 3 --nstates 15 -o parity.npz
xphd irreps --save ../03_nscf/SAVE --bse-dir ../05_bse/output --dmats Dmats.npy --nv 3 --points G K M --nstates 4
xphd band-parity --save ../03_nscf/SAVE --bse-dir ../05_bse/output --dmats Dmats.npy --nv 3 --path G M K G -o
   band_parity.npz
```

What you should see

- `parity.npz`: χ of every state at every $\mathbf{Q}$; every one of the 576×15 values lies within 7×10^{-5} of ± 1.
- The lowest band is even inside a closed contour around Γ and odd outside it, including $\mathbf{K}$ and $\mathbf{M}$ (76% of the zone is odd). The contour lies at $0.256\,|\mathbf{b}|$ along Γ–$\mathbf{K}$ and $0.265\,|\mathbf{b}|$ along Γ–$\mathbf{M}$ (`scripts/crossing_radii.py`).
- `irreps`: the lowest exciton at $\mathbf{K}$ is A″; the bright doublet at Γ is E′.

8.2 Phonon parity in the archive's mode order

```
python scripts/labels_from_elph.py      # SETTINGS: ELPH, ARCHIVE = GI_ExcPh_Q0001.npz, SAVE
```

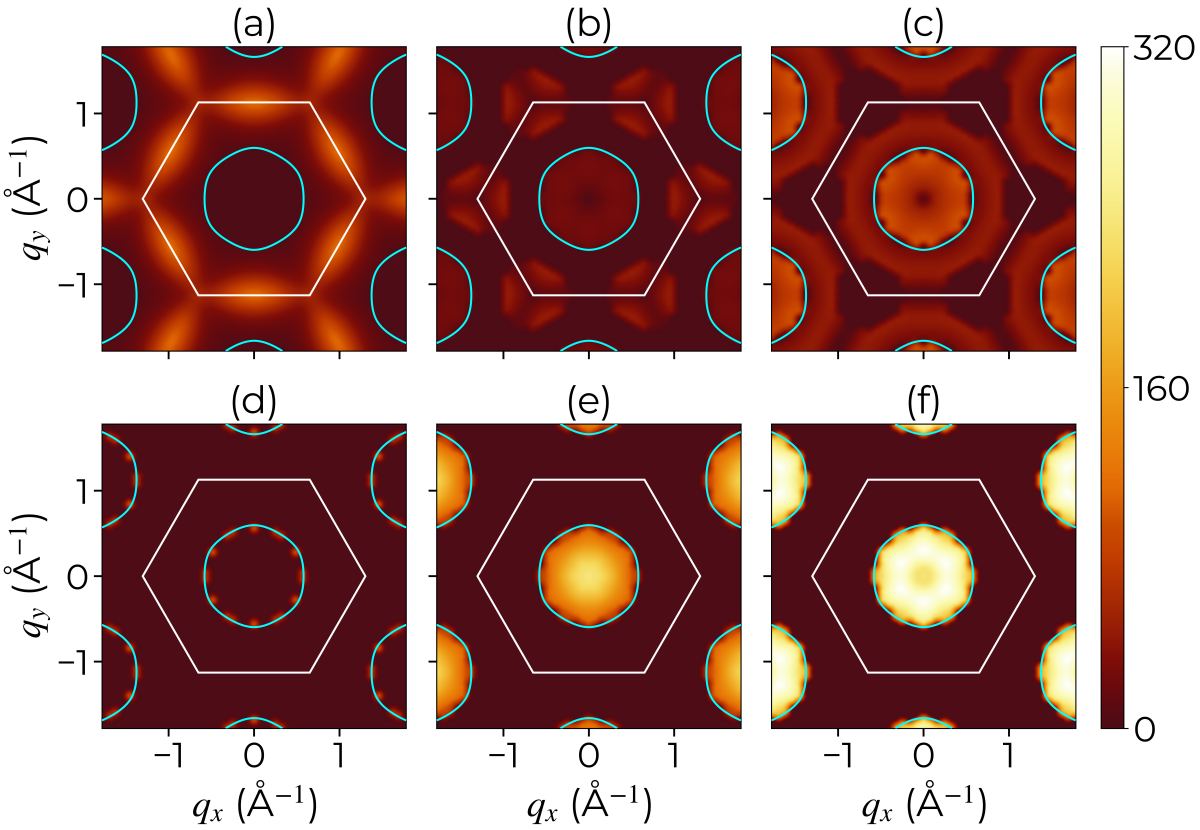
The labels come from the electron–phonon database's own eigenvectors, so they follow the archive's mode order exactly. (`matdyn` orders nearly degenerate modes differently at about 5% of the points.) Branch indices are sorted by energy at every $\mathbf{q}$, so a branch can change parity across the zone where two branches cross. The script writes `mode_labels_elph.npy` and should report a planar layer.

8.3 The rule, quantified

```
xphd selection-rule --archive archives/GI_ExcPh_Q0001.npz --parity parity.npz \
      --labels mode_labels_elph.npy --state 1 --by-final
```

What you should see

Forbidden/allowed coupling $= 5 \times 10^{-7}$: zero to numerical precision.



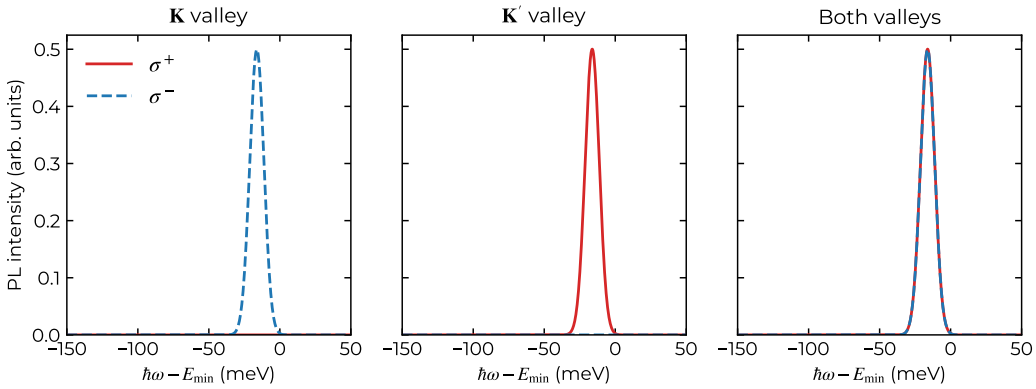
$|\mathcal{G}|^2$ (meV^2) from the bright exciton at Γ into the lowest band at $\mathbf{q}$, one panel per branch (a–f: $\nu = 1 \dots 6$). Cyan: the parity contour, inside which the lowest band is even. The polar branches (e,f) couple only inside the contour, where the final state has the bright exciton's parity; outside it they are exactly zero. These maps are drawn from `g2_grid` of the Γ archive and `parity.npz`.

9. Optics: the dark exciton's phonon sideband

The dark exciton at $\mathbf{K}$ cannot emit a photon directly: a photon carries no momentum. It can emit one together with a phonon of momentum $\mathbf{K}$. At $\mathbf{K}$ the symmetry is C_{3h}; the dark exciton is A″, the bright states E′, and E′$\otimes$A″ = E″. Of the six branches at $\mathbf{K}$, only the E″ flexural mode (16.2 meV) can make the transition, and it carries a definite circular polarisation.

```
xphd dipoles --save ../03_nscf/SAVE --bse-dir ../05_bse/output -o exc_dipoles.npy --deg-tol 0.005
python scripts/helicity_pl.py           # SETTINGS: ARCHIVE = GI_ExcPh_Q0001.npz, DIPOLES = exc_dipoles.npy
```

`xphd dipoles` contracts yambo's single-particle dipoles with *one* set of exciton eigenvectors, so the relative phase of x and y (which is what a circular polarisation is) is meaningful. Use `-deg-tol` so that degenerate states are handled as a manifold.



What you should see

- One satellite, at 3.4128 eV: 16.2 meV below the dark exciton, the energy of the E″ mode.
- Degree of circular polarisation $P = -1.0000$ for emitters in the $\mathbf{K}$ valley and $+1.0000$ in $\mathbf{K}'$ (which one is called σ^+ is a convention).
- Summed over both valleys the light is unpolarised, as time reversal requires; the two valley intensities agree to 0.12%.
- The emission comes from branch 1 alone; the other five branches contribute nothing.

If $|P|$ comes out clearly below 1 (with one of our dipole files it was 0.96), check that the dipoles were written with `-deg-tol` and from the same BSE as the archive. The polarisation lives in the relative phase of the x and y dipoles of the *same* states.

10. Real-time transport

The Boltzmann equation follows the occupation of every state at every $\mathbf{Q}$ (576 points $\times$ the states kept) in time. Its rates are the same golden-rule rates as the linewidths, assembled into a matrix that connects states at different $\mathbf{Q}$. Detailed balance is built in, and each row is rescaled so that its total equals the linewidth integrated on the fine mesh.

10.1 How many states to keep

The transport matrix grows as (states)^2. Our runs kept the lowest six states at every $\mathbf{Q}$, $576 \times 6 = 3\,456$ states; `scripts/slice_archive.py` writes copies of the archives with the first n states. Check that the result does not change when you keep more.

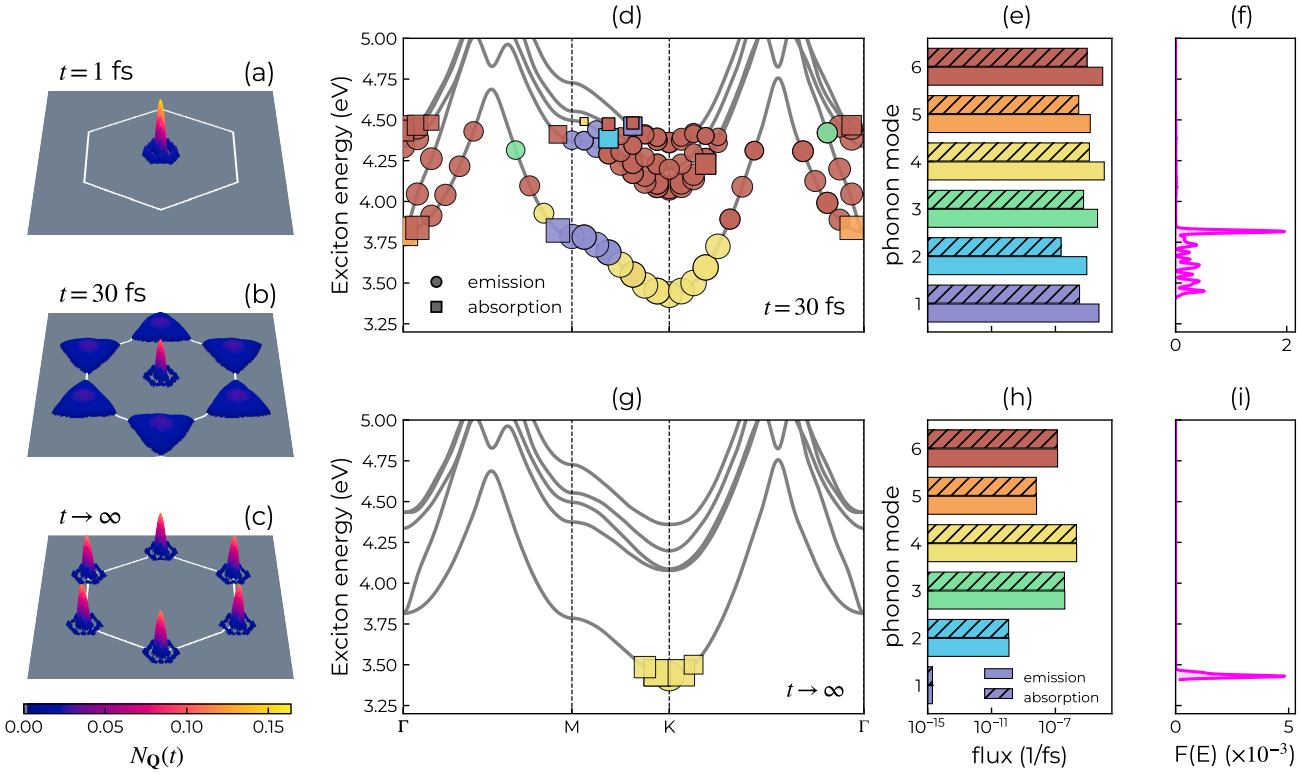
10.2 Run

```
# one node: --workers for the refinement, --threads for the propagation
xphd bte 'archives6/GI_ExcPh_Q*.npz' --T 77 --hw-fine hw_fine.npy \
      --inject optical --pump-center 3.814 --pump-sigma 0.018 \
      --t-end 10000 --nt 200 --workers 16 --threads 16 --checkpoint ck_bte \
      --channels --snapshots snaps_77K.npz
# a cluster: MPI refinement, then one node of threads (scripts/hpc/slurm_bte.sh)
python scripts/check_bte.py           # SETTINGS: BTE = snaps_77K.npz, T_BTE = 77
python scripts/valley_times.py        # SETTINGS: SNAPS = snaps_77K.npz
xphd bte-check --archives 'archives6/GI_ExcPh_Q*.npz' --field lw_field.npz --hw-fine hw_fine.npy --T 77
```

`-pump-center` and `-pump-sigma` inject resonantly into the bright doublet. Without them the injection follows the oscillator strengths of all states at $\mathbf{Q} = 0$.

What you should see

- At the start, `check_balance`: the Bose–Einstein distribution is stationary to round-off.
- **77 K**, resonant injection: the $\mathbf{K}$ and $\mathbf{K}'$ valleys hold 35% of the population at 30 fs, 50% at 41 fs, 90% at 98 fs and 99% at 172 fs. The gas settles 4.26 meV above the minimum, against 4.09 meV for a Boltzmann distribution on the same states.
- **300 K**, injection by oscillator strength: 43% in the valleys at 30 fs, 94% at 100 fs and about 99.8% at the end; the mean energy, 26.60 meV above the minimum, equals the Boltzmann value.
- `bte-check`: out-rates equal the linewidths divided by $\hbar$, ratio 1.000, when both use `hw_fine.npy`. Without the refinement the raw matrix is about 2.5 times too slow.
- The number of excitons is conserved: the run prints the drift of N/N_0, which should be negligible.



Real-time thermalisation at 77 K after resonant injection into the bright state at Γ. (a–c) Occupation over the zone at 1 fs, 30 fs and after thermalisation. (d,g) The same population on the exciton bands at 30 fs and $t \to \infty$; marker area is the occupation, colour the phonon branch that fills each state, circles emission and squares absorption. (e,h) Scattering flux through each branch. (f,i) The energy distribution $F(E)$. The population leaves Γ for the $\mathbf{K}$ and $\mathbf{K}'$ valleys within about 100 fs and stays there.

Which phonons fill the valleys. The snapshots store the flux through every branch, by emission and absorption, at every time (`flux_t`, `dom_mode_t`, `is_emission_t`). The odd flexural mode that produces the luminescence sideband is not the branch that carries the population into the valleys: the states that feed the valleys and the valley states themselves are both odd, so only an even phonon connects them.

11. When something looks wrong

Symptom	Cause and fix
`check-archive`: one symmetry far above the other	a rotation fault in the Dmats/BSE chain; or, with `-detail`, a split doublet: raise `-deg-tol`
Acoustic $\|\mathcal{G}\|^2$ at Γ not zero	wrong relative sign of the electron and hole terms
A spurious ≈15 meV LO–TO splitting at Γ	`loto_2d=.false.` somewhere in `q2r.x`/`matdyn.x`
Exciton minimum near 3.355 eV instead of 3.429 eV	a database from the buckled cell
Linewidth at a band minimum is zero	the default acoustic model drops small-$\mathbf{q}$ absorption; use `-acoustic-model q2`
$\|P\|$ of the sideband clearly below 1	dipoles without `-deg-tol`, or from another BSE
Transport out-rates 2–3 times too slow	no refinement: pass `-hw-fine`
Population vanishes into one $\mathbf{Q}$	an archive is missing (`verify-archives`)
Cartesian quantities slightly off	a script left at its default lattice constant (3.2 Å); pass 3.21 Å

A. Every command, in order

```
# 1. ground state and phonons (QE)
pw.x -in scf.in > scf.out                       # after vc-relax to the planar cell
ph.x -in ph.in > ph.out                         # 24x24x1, tr2_ph = 1e-18
q2r.x < q2r.in > q2r.out                        # zasr = 'crystal', loto_2d = .true.
xphd matdyn-split --n 360 --chunks 54
sbatch scripts/run_matdyn.sh                    # matdyn.x on every chunk, merged
xphd modes-check --modes matdyn.modes --freq gan.freq --n 360

# 2. quasiparticles and excitons (yambo), electron-phonon (LetzElPhC)
pw.x nscf  ->  p2y  ->  yambo                   # SAVE
yambo  GW0  ->  BSE at all 61 irreducible Q     # ndb.BS_diago_Q1..61, 15 states
LetzElPhC                                       # ndb.elph, from the same SAVE

# 3. archives
xphd dmats --bse-dir BSE --savepath SAVE --ndb-elph ndb.elph --nexc 15
xphd generate 0 archives --nexc 15 --mesh 24 24 --savepath SAVE --ndb-elph ndb.elph --bse-dir BSE --dmats Dmats.
   npy
xphd check archives/GI_ExcPh_Q0001.npz
xphd check-archive archives/GI_ExcPh_Q0001.npz                    # READY
xphd generate 1-575 archives --resume ... (same options)          # or scripts/hpc/slurm_generate.sh
xphd verify-archives --dir archives --n 576
xphd matdyn gan.freq --fine 360 --nbnd 6 --npz archives/GI_ExcPh_Q0001.npz -o hw_fine.npy

# 4. linewidths
xphd linewidth archives/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 --acoustic-
   model q2 --n-shell 1 --per-mode -o G.txt
xphd linewidth archives/GI_ExcPh_Q0575.npz --hw-fine hw_fine.npy --T 5 77 300 --acoustic-cut 5e-4 --acoustic-
   model q2 --n-shell 1 --per-mode -o K.txt
xphd channels  archives/GI_ExcPh_Q0001.npz --state 1 --T 77
xphd sweep 'archives/GI_ExcPh_Q*.npz' --hw-fine hw_fine.npy --T 5 10 20 40 77 100 150 200 250 300 --acoustic-cut
   5e-4 --acoustic-model q2 --n-shell 1 --workers 8 --unfold SAVE --checkpoint ck_sweep -o lw_field.npz

# 5. symmetry
xphd parity --save SAVE --bse-dir BSE --dmats Dmats.npy --nv 3 --nstates 15 -o parity.npz
xphd irreps --save SAVE --bse-dir BSE --dmats Dmats.npy --nv 3 --points G K M --nstates 4
xphd band-parity --save SAVE --bse-dir BSE --dmats Dmats.npy --nv 3 --path G M K G -o band_parity.npz
python scripts/labels_from_elph.py                                 # mode_labels_elph.npy
xphd selection-rule --archive archives/GI_ExcPh_Q0001.npz --parity parity.npz --labels mode_labels_elph.npy --
   state 1 --by-final

# 6. optics
xphd dipoles --save SAVE --bse-dir BSE -o exc_dipoles.npy --deg-tol 0.005
python scripts/helicity_pl.py

# 7. transport
xphd bte 'archives6/GI_ExcPh_Q*.npz' --T 77 --hw-fine hw_fine.npy --inject optical --pump-center 3.814 --pump-
   sigma 0.018 --t-end 10000 --nt 200 --workers 16 --threads 16 --channels --snapshots snaps_77K.npz
python scripts/check_bte.py
python scripts/valley_times.py
xphd bte-check --archives 'archives6/GI_ExcPh_Q*.npz' --field lw_field.npz --hw-fine hw_fine.npy --T 77
```

B. The files, and what is inside

File	Contents
`Dmats.npy`	rotation matrices, (symmetry operations, $\mathbf{k}$, bands, bands)
`GI_ExcPh_Q`*nnnn*`.npz`	one $\mathbf{Q}$: `G_grid`, `Ge_grid`, `Gh_grid` (complex, eV) and `g2_grid`, each ($\mathbf{q}$, branch, initial, final); `E_n_grid`, `E_m_grid`, `hw_grid`, `q_red`, `Q_red`, `mesh`
`hw_fine.npy`	phonon energies on the 360×360 mesh, eV, $(129\,600, 6)$
`lw_field.npz`	`LW` (T, $\mathbf{Q}$, state) in eV, `T`, `Q_red`, `E_n`, `mode_em`, `mode_ab`
`parity.npz`	`chi` ($\mathbf{Q}$, state), `E`, `Q_red`
`mode_labels_elph.npy`	parity of every ($\mathbf{q}$, branch) in the archive's order
`exc_dipoles.npy`	complex x, y, z dipoles of the $\mathbf{Q} = 0$ states, (3, states)
`snaps_77K.npz`	`t`, `F` (time, state), `E`, `Q_red`, `band`; with `-channels` also `flux_t`, `dom_mode_t`, `is_emission_t`, `dom_frac_t`